

Farzaneh Heidari

- farzaneh.heidari@mila.quebec
 - [LinkedIn](#) • [GitHub](#) • [Publications](#)
-

Education

Universite de Montreal (Mila)	MONTREAL, CANADA
PhD. in Computer Science (Interpretability and Tensor Networks)	2021–
Advisors: Guillaume Rabusseau, Jian Tang	
projects: Tractable Shapley Values and Nash equilibrium, Evaluation Awareness in LLMs, Multi-linear Steering	
York University	TORONTO, CANADA
M.Sc. in Computer Science (Graph Representation Learning)	2017– 2019
Advisor: Manos Papagelis	
overall GPA: 4.06/4	
Sharif University of Technology	TEHRAN, IRAN
B.S. in Electrical Engineering (Telecommunications), Minor in Economics	2012– 2017
overall GPA: 16.28/20	
Farzanegan High school	
National Organisation for Development of Exceptional Talents (NODET)	SANANDAJ, IRAN
Pre-university Certificate and High School Diploma in Mathematics & Physics	2008– 2012
GPA: 19.79/20	

Employment

Bioinformatics lab at Université du Québec à Montréal and IRDA	MONTREAL, CANADA
Machine Learning Researcher	Dec 2019- May 2021
• Causal Inference for climate change impacts on agriculture	
In this project, we developed machine learning algorithms to predict physical characteristics of soil and designed interpretable and causal models to provide insight into the predictions, in collaboration with agriculture and soil experts. The results were published in IRDA internal reports.	

Research Experience

LawZero	Montreal, Canada
AI Safety	February 2026- present
AI safety and evaluations using mechanistic interpretability tools.	
Big data Research, Analytics, and Information Networks (BRAIN) Lab	Toronto, Canada
Graph Representation Learning	September 2017- September 2019
Under the supervision of Prof. Manos Papagelis, I worked on Graph Representation Learning. My main focus was learning evolving graph representations and trajectory networks.	
The Mar Lab	Toronto, Canada
Deep Learning on Narrative Fiction Data	February 2018- September 2019
Under the supervision of Prof. Raymond A. Mar, I collaborated on network representation of narrative fiction data. Mostly, diffusion of influence in the emotional reactions to the fictional couples.	
Ambient Intelligence Research (AIR) Lab	Tehran, Iran
Pattern Recognition	September 2015-June 2016
Under the supervision of Prof. Aghajan, I worked on extracting behavioral patterns in smart environments. Our main focus was using deep learning in public transportation, efficient buildings, interactive wallpapers and stairs.	
Graduate School of Business and Economics (GSME), Sharif University of Technology Tehran, Iran	
Regression Analysis	August 2013-April 2014
Under the supervision of Prof. Rahmati, I analyzed Iran's trade datasets and interpreted the data using linear regression analysis.	

Internship

Arya Hamrah

Tehran, Iran

Three months summer internship (Telecommunication Network)

June 2014–September 2014

As a two-member team, we were responsible for developing an extension for Android Mobile Application of Irancell (Iranian telecommunications company) MTN's DataCenter; My role mainly was Java programming and monitoring the communication network overload.

Helmholtz-Institute for Biomedical Engineering RWTH Aachen University

Aachen, Germany

Five months internship (Biomedical Engineering)

July 2016–December 2016

As an intern, I worked in risk management and usability research group and was responsible for integrating high-frequency devices to the operation room under supervision of Dr. Ing. Armin Janss and Prof. Radermacher.

Samsung Research America

Montreal, Canada

Four months Mitacs internship (Graph Explainability)

September 2022–January 2023

Under supervision of [Perouz Taslakian](#), I worked on explainability methods for graph neural networks (GNN), the result of the internship was a submission to Topology, Algebra and Geometry workshop at ICML 2023. This is an ongoing research.

Zapata Computing AI

Montreal, Canada

Five months Mitacs internship (Stable Diffusion Explainability)

June 2024–October 2024

Under supervision of Jing Chen and Jacob Miller I worked on explainability methods for stable diffusion models, more specifically I worked on controlled generation.

RIKEN AIP

Tokyo, Japan

Three months internship (Tractable calculation of Shapley interactions)

July 2025–October 2025

Under supervision of Chao Li, I worked in the tensor learning team and developed tractable game theory index calculation algorithms for explainability in deep learning models.

Summer School and Workshop

Deep Learning and Reinforcement Learning Summer School 2019 (*CIFAR & Amii*) Edmonton, Canada

Mechanism Design Workshop (Sharif University, Fall 2015) by [Prof. Arunava Sen](#). Tehran, Iran

Publications

Conference

Heidari, Farzaneh, Rabusseau, Guillaume . "TN-SHAP-G: Graph-Structured Tensor Network Surrogates for Shapley Values and Interactions " ICML 2026.

Conference

Heidari, Farzaneh, Li, Chao, Rabusseau, Guillaume . "Tractable Shapley Values and Interactions via Tensor Networks" AISTATS 2026.

Workshop

Heidari, Farzaneh, Taslakian, Perouz and Guillaume Rabusseau. "Explaining Graph Neural Networks Using Interpretable Local Surrogates." Proceedings of 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML), PMLR 221:146-155, 2023.

Conference (Oral Presentation)

Heidari, Farzaneh, and Manos Papagelis. "EvoNRL: Evolving Network Representation Learning Based on Random Walks." International Conference on Complex Networks and their Applications. Springer, Cham, 2018.

Conference

G. Alix, N. Yanin, T. Pechlivanoglou, J. Li, Farzaneh. Heidari and M. Papagelis."A Mobility-based Recommendation System for Mitigating the Risk of Infection during Epidemics" 2022 23rd IEEE International Conference on Mobile Data Management (MDM), Paphos, Cyprus, 2022, pp. 292-295, doi: 10.1109/MDM55031.2022.

Journal

Heidari, Farzaneh, and Manos Papagelis. "Evolving Network Representation Learning Based on Random Walks." Applied Network Science (Machine Learning with Graphs).

Journal

Pechlivanoglou. T, Li, J, Sun, J, Heidari, Farzaneh and Papagelis, Manos."Epidemic Spreading in Trajectory Networks " Big Data Research, Volume 27, 2022,

Short Paper

Raymond A.; van Dalen–Oskam, Karina; Temple, Emily; Bowman, Isabel; Heidari, Farzaneh; Nguyen, Ahn T.P. "Features of Timelessness: Intermediate Report on a Quest for Stylistic Features that Mark Literary Canonicity" Digital Humanities 2020.

BSc. Thesis

Study and Analysis of Wholesale Electricity Markets

Predicting the wholesale electricity market's revenue in case of the pay-as-bid pricing strategy in the auctions using STATA and regression models in MATLAB.

MSc. Thesis

Evolving Network Representation Learning based on Random Walks¹

To extend static random walk based network representation methods to evolving networks, we proposed a general framework for updating random walks. The updated random walks leverage time and space efficiency of inverted indexing methods. We discussed the trade-off between time performance and accuracy when obtaining concurrent network representations. To determine the right time for obtaining a network embedding, we demonstrated that simple strategies for monitoring the changes that occur in the network can provide support in decision making.

Selected Courses

Online: QxQ - Introduction to Quantum Computing by IBM Quantum

Mila (ML): Probabilistic Graphical Models, Matrix and Tensor Factorization Techniques for Machine Learning

York University (ML): Machine Learning Theory (prof. Ruth Urner), Data Mining, Data Analytics and Visualization, Probabilistic Models & Machine Learning, Graph Representation Learning.

Sharif University of Technology (Telecommunications): Communication Systems, Data Structure and Algorithms, Fundamentals of Biomedical Engineering, Signals & Systems, Data Networks, Digital Communications, Digital Signal Processing, EM Fields & Waves, Wireless Communications Lab (SP Applied Issues in Transcripts), RF Communications & Circuits Lab, Waves & Antennas, Game Theory (Ph.D.)

Teaching Assistance

York University: Introduction to Machine Learning and Pattern Recognition (prof. Ruth Urner, *Wi 2019*), Computing for Psychology (*Wi 2018*), Systems and Random Processes (*Fa 2017*), Net-centric Introduction to Computing (*Fa 2017*)

Sharif University of Technology: Econometrics (*Wi 2015*)

Volunteer Experience

- Organizing Tensor Network Reading Group September 2023- Present

Founded and coordinate a weekly research reading group focused on tensor networks, interpretability, and related machine learning methods. Responsibilities include selecting papers, preparing discussion material, moderating sessions, and facilitating collaboration among graduate students and researchers (approximately 1–2 hours per week).

- Member of sustainability committee at Mila. January 2022- Present

Active member of Mila's sustainability committee, participating in regular meetings and working groups focused on reducing the environmental impact of research activities. Contributed to the design and organization of hybrid and virtual conferences to enable participation without travel, and to decision-making processes on sustainability initiatives (approximately 2–3 hours per month).

- Member of sustainability group at Mila. December 2021- December 2022

Joined the sustainability group at Mila during its initial phase and contributed to defining its direction and activities. Participated in early planning meetings, discussions on carbon accounting, and the development of sustainable research practices that later fed into the formal Sustainability Committee (approximately 2 hours per month).

- Teaching extracurricular Physics to high school students at Yarigaran charity foundation of Sharif University. September 2014- April 2015

¹code available in <https://github.com/farzana0/EvoNRL>

Honors and Awards

- Ranked **35th** among more than **250,000⁺** participants in Nationwide University Entrance Exam for B.Sc. degree, 2013.
 - Semifinalist in national olympiad in **Mathematics** and **Physics** , 2010.
 - Bronze medalist in national chess competitions among NODET students, 2007.
-

Skills

- Programming Languages: Python, C, C++, MATLAB.
- Simulation & Verification: MATLAB & Simulink.
- Tools: Networkx, PyTorch, TensorFlow, ElasticSearch, Gephi, Cloud Computing, Causal Inference libraries, RL, iTensor

Natural languages: Kurdish (*Native*), Persian (*Native*), English (*Full Professional Proficiency*), French (*Upper Intermediate*)